What Prices Cannot Tell You: An Identification Framework for Information Transmission in Live Markets

Markets reveal prices. They do not reveal how those prices came to be.

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Working Paper · DRAFT, Sections 1-5 · Results not yet written

> **Draft status.** This manuscript is complete through the Methods section. The Results and Discussion sections are deliberately unwritten: they are gated on four objective measurement conditions stated in Section 5.6, and the analysis plan in Section 5.5 is pre-registered so that the design cannot be shaped by the evidence. Nothing in the sections below reports an empirical finding from the dataset described in Section 4.

Abstract

A companion study found that a live baseball wagering market encompasses every public-information candidate tested against it: no variable in a ten-hypothesis ladder improved on the market's own forecast of remaining runs. That null is a statement about outcomes, and it invites a question about mechanism. If the market has already incorporated public information, when did it do so, and is the apparent efficiency genuine or partly an artifact of observing the market only after the information has already propagated?

We take up the transmission question directly, and find that it is largely an identification problem. The delay between a game event and an observed price change decomposes into a bookmaker's pricing latency, its feed's transport latency, and the observation latency imposed by our own sampling. That decomposition is arithmetic. The substantive difficulty is that only their sum is observed, and this paper is about what can and cannot be learned under that constraint. We show that three materially different markets, one in which a bookmaker genuinely prices faster, one in which pricing speeds are identical and only publishing infrastructure differs, and one in which both differ and partially offset, generate numerically identical timestamp data. No statistic computed from timestamps alone separates them.

Working from a two-book, thirty-one-second live quote panel matched to game-state events, we define an event-anchored response latency and state precisely the conditions under which its cross-book contrast is a statement about pricing rather than about plumbing. We show that several intuitive estimators of information leadership are mechanically confounded by update frequency and by the analyst's choice of which posted line to treat as the main line, and we characterize the resolution floor below which this class of instrument is silent. The design admits three outcomes, all of which we regard as publishable: identification of a pricing contrast, identification of bounds only, or a demonstration that the quantity is not identifiable without richer instrumentation. The third is not a failure of

the study. Empirical work in this area has generally assumed identification rather than established it, and showing where the assumption breaks is itself a contribution.

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1. Introduction

Every pitch thrown in a Major League Baseball game can trigger dozens of price updates across sportsbooks before the next pitch is delivered. These are high-frequency markets in the ordinary sense of the term, and they have an unusual property that makes them a good laboratory for studying how prices form: unlike an equity, every contract settles within hours against an objective terminal payoff, so the thing being forecast is eventually revealed and the forecast can be graded.

Market efficiency tests answer a question about outcomes. They ask whether some information, held by someone, would have improved on the price. When the answer is no, as it frequently is, the finding is usually reported and the matter closed. But a null of that kind is a statement about a mechanism whose operation has not been observed. Something incorporated the information. The efficiency test does not say what, or how quickly, or through which participant.

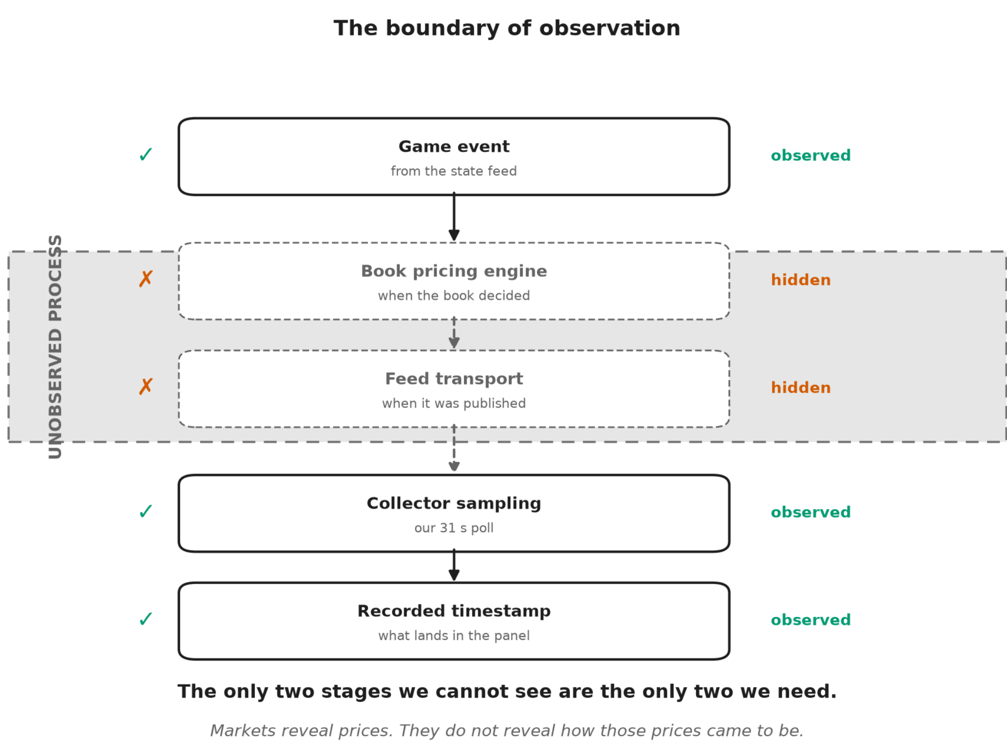

Figure 1. *We never observe the pricing process itself.* Of the five stages between a game event and a row in our dataset, the two in the shaded band are exactly the two the question is about: when the bookmaker decided to move, and when its feed made that decision visible. Everything we record sits below the boundary.

This paper begins where a companion study ended. That study treated a sharp live betting line as an incumbent forecast of the runs remaining in a baseball game and asked whether any of ten publicly observable candidates, pitcher fatigue and velocity decay among them, carried incremental predictive content beyond it. None did. The market's forecast error was unpredictable from every variable measured, and the one candidate that appeared alive was shown to be a post-treatment selection artifact rather than an effect. The conclusion was a boundary: the frontier of public information had already been reached by the price.

The transition between the two studies can be stated in a sentence. **The companion study established that public baseball variables contain no incremental information beyond a sharp market observed at one-minute resolution; this paper asks whether that apparent efficiency is genuine, or whether it is partly an artifact of observing the market only after the information has already propagated through it.** The companion study's own closing section anticipates the question, listing cross-book propagation, the information half-life of a shock, and the microstructure limits of one-minute single-book snapshots as the things its data could not settle.

The natural next question is not whether the boundary exists but how it is enforced. A live betting market is an unusually good laboratory for that question. Unlike an equity price, whose fundamental value is never realized and whose information arrivals are diffuse and hard to timestamp, a baseball market is punctuated by discrete, publicly observable, precisely dated events. A run scores. An out is recorded. A pitcher is removed. Each is an information arrival with a known clock time, and each is followed, sooner or later, by a revised price. The interval between the two is the object of this paper.

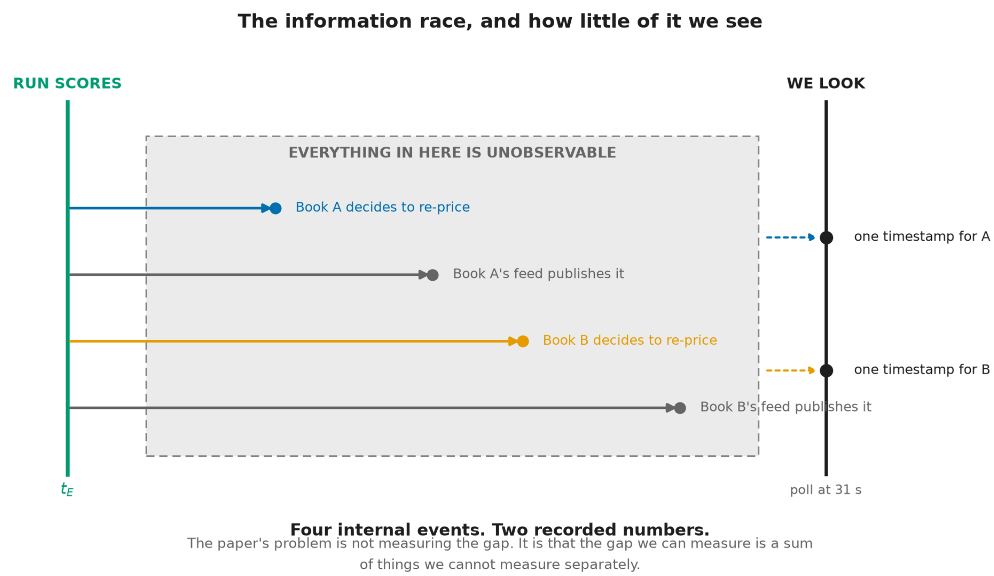

Figure 2. *Four internal events; two recorded numbers.* Between the run and our record of a new price sit each book's decision to re-price and each book's publication of that decision. We hold two timestamps. Between the run and our record of a new price sit four internal events, none of which we see: each book's decision to re-price, and each book's publication of that decision. We hold two timestamps. The difficulty of this paper is not measuring the interval; it is that the interval we can measure is a sum of intervals we cannot measure separately.

Box 1. A thought experiment

Two sportsbooks are quoting the same game. A run scores. Book A updates its price 800 milliseconds later; Book B updates 1.6 seconds later. You observe both prices once every 30 seconds.

Can you conclude that Book A processes information faster?

No. You did not observe the pricing engines, the feed delays, or the moment the event actually occurred. You observed two timestamps, produced by a sampling process of your own construction, at a resolution nearly twenty times coarser than the difference you are trying to detect. Both books land in the same poll. The data are silent.

This paper is about what it takes to make them speak, and about being candid when they cannot.

Studying that interval requires more than one price. A single book's response time is not interpretable on its own: it confounds how fast the bookmaker thinks with how fast its infrastructure publishes and how fast the analyst happens to be looking. Comparing two books holds out the possibility that the shared components cancel. Whether they actually do is an empirical and architectural question, and answering it honestly is the substance of what follows.

The problem has a familiar shape. Astronomers infer the existence and mass of a planet they cannot see from the wobble of a star they can. The inference is only as good as the model of what lies between the observer and the object, and a great deal of the work is in characterizing the instrument rather than the sky. Our position is the same: we infer an invisible pricing process from the visible prices it produces, and the difficulty is not the arithmetic but the intervening machinery. **Markets reveal prices. They do not reveal how those prices came to be.**

We therefore frame this as a paper about identification in high-frequency markets rather than a second efficiency paper, and rather than a paper about sportsbooks. Sports betting is the laboratory, chosen because its information arrivals are discrete and precisely dated and its contracts settle against ground truth; the identification problem it exposes is general. The question is not whether the market can be beaten. It is how information moves through a market that apparently cannot be, and whether the movement is measurable with instruments of the kind available to an outside observer. Three things follow from taking that question seriously.

First, the estimand must be defined before the data are examined, because the intuitive estimators are wrong in a specific and instructive way. The obvious approach, timing one book's price change against the other's, is mechanically biased toward whichever book updates more often. A book that re-prices every thirty seconds will reach any given price level before a book that re-prices every eight minutes, regardless of which one is better informed. That is not leadership; it is sampling.

Second, the identification problem must be confronted rather than assumed away. The delay we observe is a sum of three delays, and only one of them is about pricing. Separating them requires either an independent measurement of transport latency or an argument that transport latency is common across books. Neither is free, and we do not assume either.

Third, the instrument's limits must be stated in advance. A panel sampled every thirty-one seconds cannot resolve sub-second price formation. If the economically interesting action happens below the sampling floor, no amount of care in the analysis will recover it, and the honest report is that the question is out of reach for this apparatus.

The paper is organized around those three commitments. Section 2 develops the conceptual framework and decomposes the observed delay. Section 3, the heart of the paper, treats identification: what can and cannot be learned from a two-book panel, and under exactly which assumptions. Section 4 describes the data and explains why this instrument differs fundamentally from the one used in the companion study, a difference that precludes any direct comparison of results. Section 5 states the estimation procedure and the pre-registered analysis plan, and closes by listing the objective conditions that must be satisfied before results may be reported.

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2. Conceptual framework

2.1 The transmission chain

Consider a discrete, publicly observable event in a baseball game occurring at clock time `t_E`: a run scores. The event changes the conditional distribution of the game's final total, and a bookmaker offering a live total on that game will, at some point, revise its posted number. An outside observer records the revision at some later time. Between the event and the record lie three distinct delays, produced by three distinct mechanisms.

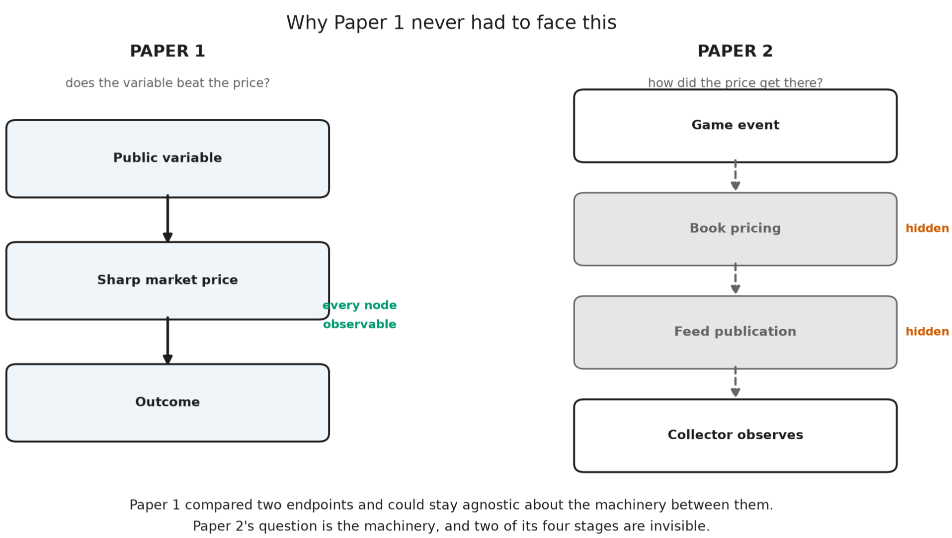

Figure 3. *Paper 1 never needed to identify internal latency.* It compared two endpoints with the market between them, and every node it required was observable. This paper's question is the machinery itself, and half its stages are hidden. It compared two endpoints, a public variable and a realized outcome, with the market price between them, and every node it needed was observable. This paper's question is the machinery itself, and two of its four stages are hidden from any outside observer.

Pricing latency, which we write as the interval between the event and the bookmaker's internal decision to revise, is the economically meaningful quantity. It reflects how quickly the bookmaker detects the event, updates its model, and commits to a new number. Differences in pricing latency across books are differences in how the market processes information.

Feed latency is the interval between the bookmaker's internal revision and the moment that revision becomes visible on the public endpoint we query. It is a property of publishing infrastructure, caching, and content delivery, not of judgment about baseball.

Observation latency is the interval between publication and our sampling of it. It is a property of our own collector. With a polling interval of roughly thirty-one seconds, a revision published immediately after a poll waits, on average, about fifteen seconds to be seen, and up to thirty-one seconds in the worst case.

The data contain only the sum. For book `b` and event `E`, the observable is

```
Δt_b(E) = λ_price_b(E) + λ_feed_b(E) + λ_obs_b(E)
```

As an accounting identity this is unremarkable, and we do not present it as a contribution. Writing a delay as a sum of its parts is bookkeeping. What matters is the consequence: **no manipulation of a single book's series recovers the individual terms, because only the left-hand side is ever observed.** Everything difficult about this paper follows from that sentence, which is why the identification section is longer than the estimation section.

2.2 Why two books might help

If the second and third terms were identical across books, the cross-book difference would remove them:

```
Δt_A(E) - Δt_B(E) = λ_price_A(E) - λ_price_B(E)
```

and the contrast would isolate the economics. This is the hope that motivates a two-book design, and it is an assumption, not a fact.

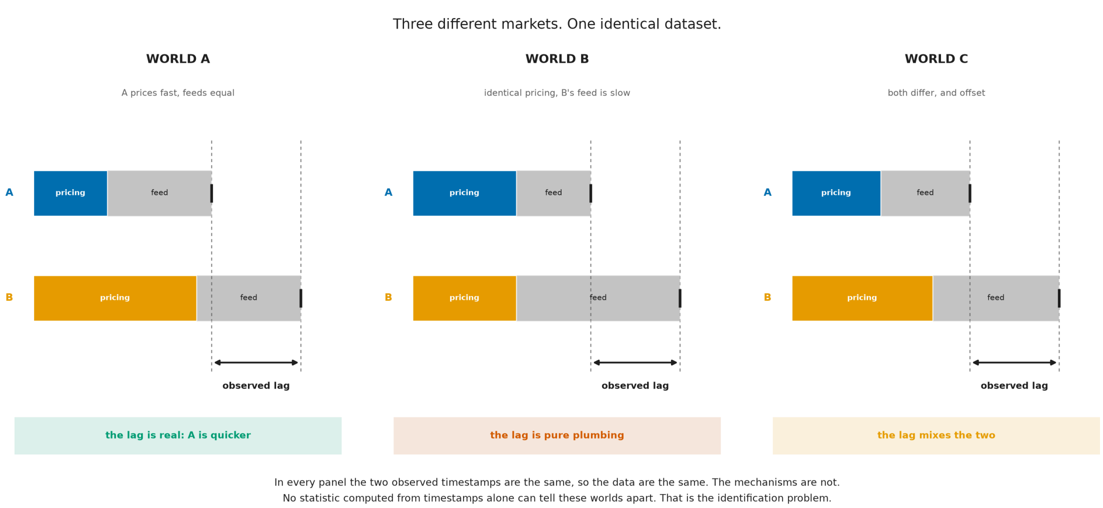

Figure 4. *Identical observations can arise from different underlying markets.* In the first world a bookmaker genuinely prices faster and the publishing infrastructure is equal, so the observed lag means what it appears to mean. In the second, the two bookmakers price at identical speed and the entire observed lag is produced by one book's slower feed. In the third, both differ and partially offset. The observed timestamps, marked by the dashed guides, fall in exactly the same places in all three panels. The datasets are numerically identical; the markets are not. Any statistic computed from timestamps alone assigns the same value to all three, so distinguishing them requires evidence from outside the timing series.

Observation latency is plausibly common-mode by construction: a single collector polls both books on the same schedule, so the sampling penalty is drawn from the same distribution for each. Feed latency is not. Two commercial sportsbooks run different infrastructure, and there is no reason in principle for their publishing delays to match. Whether the difference is material relative to the pricing differences we hope to detect is precisely what Section 3 must establish.

2.3 Why the estimand is anchored to the event

A tempting alternative avoids the event entirely: observe when each book arrives at a given price level and call the earlier one the leader. This is mechanically defective.

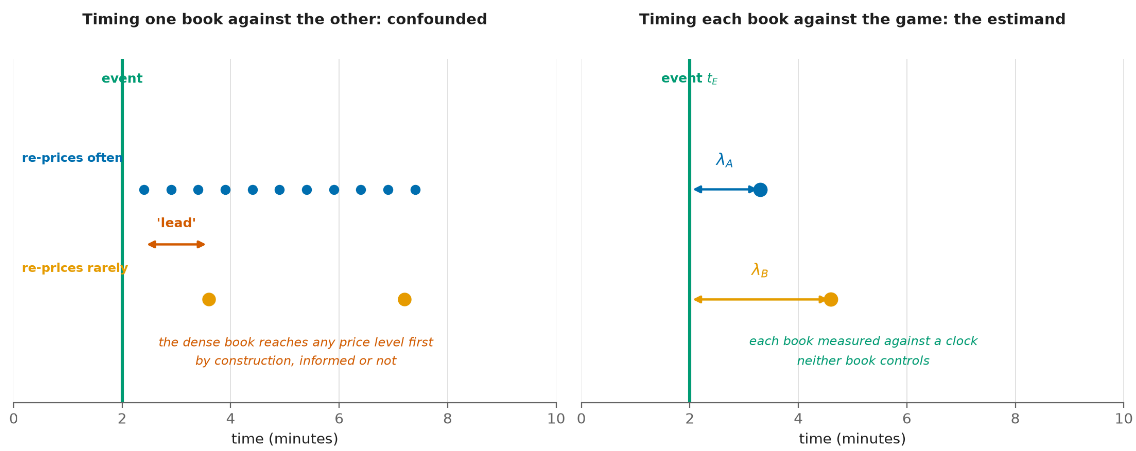

Figure 5. *A denser book leads by construction, not by insight.* Left: a book that re-prices frequently passes through any given price level before a book that re-prices rarely, purely as a consequence of update density. A leadership statistic built on book-to-book arrival times reports this sampling artifact as information leadership. Right: anchoring each book's response to the exogenous game event measures each against a clock that neither book controls, so update density no longer confers a spurious advantage.

The game event is exogenous to both books in the relevant sense: runs are not caused by bookmaker quoting behavior. Anchoring to it converts a comparison between two endogenous, differently sampled series into two comparisons against a common external clock. This does not solve the feed-latency problem, but it removes the frequency confound, which is a distinct and more easily made error.

2.4 What the market is quoting

One structural fact about the instrument shapes the analysis and is easy to get wrong. The posted live number is a **full-game total**, not a forecast of remaining runs: it is the expected final combined score, incorporating runs already scored. Consequently a run that scores enters the posted number with a positive sign, partially offset by the reduction in scoring opportunities that remain. A revision following a run is therefore expected to be upward, and its magnitude is a pass-through fraction rather than an elasticity. We verify this property directly in Section 4 rather than assuming it, because the opposite convention would reverse the sign of every response we measure.

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3. Identification

This section is the paper's core contribution. We state the assumptions under which the cross-book contrast identifies a pricing difference, examine each against what is known about the instrument, and describe what remains estimable when an assumption fails.

3.1 The estimand

Let `E` index discrete game events with clock times `t_E`, and let `t_b(E)` denote the time of book `b`'s first main-line revision attributable to `E`. Define the event-anchored response latency

```
λ_b = E[ t_b(E) - t_E ]
```

and the cross-book contrast `Δλ = λ_A - λ_B`. The target of inference is `Δλ`, and the question the paper asks is under what conditions `Δλ` is a statement about pricing rather than about plumbing.

3.2 The assumptions

A1. Event exogeneity. Game events are not caused by bookmaker quoting behavior. This is uncontroversial in this setting and is the reason event anchoring is available to us at all.

A2. Clock comparability. Event timestamps and quote timestamps are on a common, comparable clock. This is not automatic. Event times are derived from one upstream provider and quote times from another, and either may be a publication time rather than an occurrence time. A2 requires a direct audit of timestamp provenance, and any residual skew must be quantified rather than assumed away, because a constant skew shifts both `λ_A` and `λ_B` equally but a book-specific skew is indistinguishable from a pricing difference.

A3. A well-defined main line. The response `t_b(E)` presupposes a single price series per book. Books post multiple simultaneous total lines, a main line and alternates, without a discriminator. Reasonable rules for extracting the main line disagree with one another on a large majority of observations, and downstream statistics move materially depending on the choice. A3 therefore requires a fixed, documented, and tested extraction rule, together with a demonstration that the conclusions do not depend on it.

A4. Separability of transport from pricing. This is the binding assumption. `Δλ` identifies the pricing contrast only if feed latency is common-mode across books, or if a book-specific transport component can be independently measured and removed. We do not assume A4 holds. Establishing whether it holds, or demonstrating rigorously that it cannot be established with instruments of this class, is the paper's principal empirical task.

A5. Sufficiency of two books. With two books, a discrepancy is detectable but not attributable: if one book behaves anomalously there is no third observation to adjudicate which is anomalous. A5 therefore concerns robustness rather than point identification, and it is either satisfied by adding a third live source or addressed by an explicit outlier-detection procedure that does not require one.

A6. Non-arbitrary attribution. The mapping from an event to "the revision caused by it" requires an attribution window. A window chosen after inspecting the data is a researcher degree of freedom capable of manufacturing an effect. A6 is satisfied by pre-registration, which is why the window is fixed in Section 5.5 before any estimate is produced.

3.3 Three admissible outcomes

The design admits exactly three outcomes, and we commit in advance to reporting whichever obtains.

Outcome A. The pricing contrast is identified. The clock audit passes, feed latency is shown to be common-mode or is independently measured, and the extraction rule is shown not to drive the result. The cross-book contrast is then a statement about how two bookmakers process information.

Outcome B. Only bounds are identified. Some assumption holds partially: transport latency is bounded but not known, or the result is directionally stable but magnitude-sensitive to the extraction rule. The reportable object is then an interval within which the pricing contrast must lie, together with the assumption that produced it.

Outcome C. The quantity is not identifiable with this class of instrument. No external measurement of feed latency is obtainable from public endpoints, and no argument establishes common-mode behaviour. The reportable result is then a demonstration, not a guess: a proof that timestamp data from public sportsbook endpoints cannot separate market behaviour from publishing infrastructure, together with a specification of the additional instrumentation that would be required.

We want to be explicit that Outcome C is a result and not a failure. Empirical work on information transmission has generally enjoyed exchange-timestamped data and has therefore been able to assume away the decomposition this paper confronts. Where that assumption is unavailable, the discipline is to demonstrate its unavailability rather than to proceed as though it held. A paper that establishes where identification breaks down tells a subsequent researcher what to build.

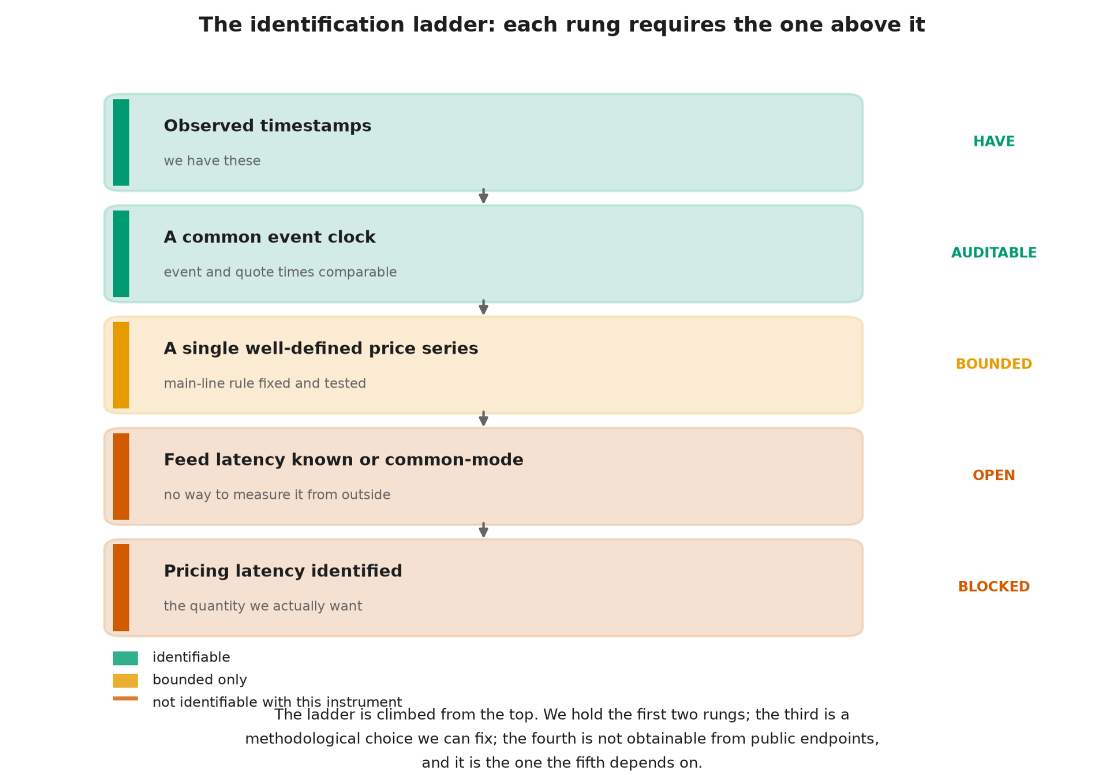

Figure 6. *Identification requires assumptions, not computation.* The requirements form a ladder, and each rung depends on the one above it. Each rung depends on the one above it. We hold the top two. The third is a methodological choice we can fix and test. The fourth, knowledge of feed latency, is not obtainable from public endpoints, and the fifth depends on it. The colour of the fourth rung is what determines whether this paper reports Outcome A, B, or C.

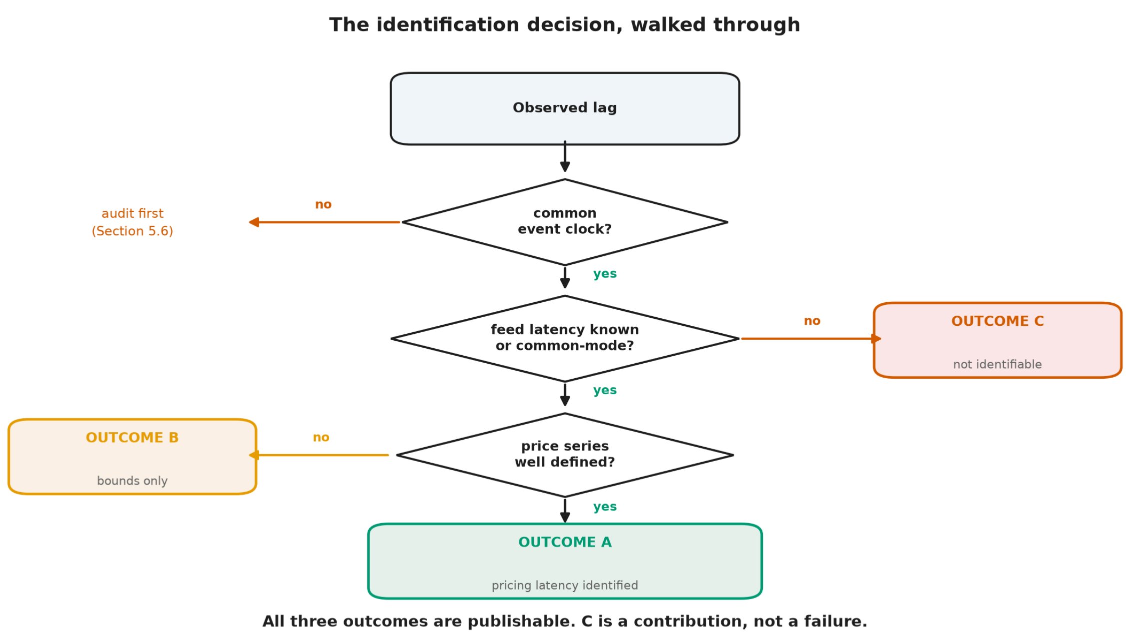

Figure 7. *The paper does not presuppose which branch it takes.* The same logic, as a decision the reader can walk. The paper does not presuppose which branch it will take. All three terminal states are publishable, and the third is the one the literature most often skips.

3.4 The resolution floor

An instrument cannot report structure below its sampling interval.

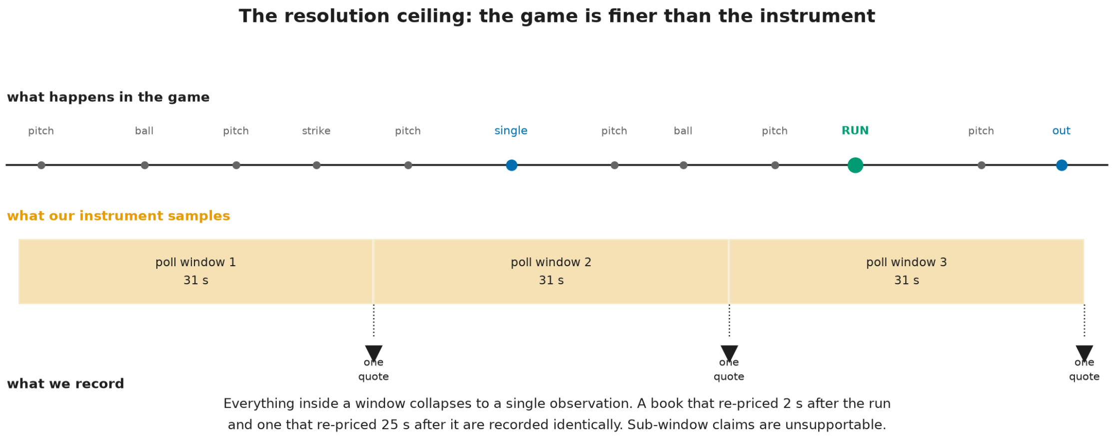

Figure 8. *The instrument itself limits what can be learned.* The game is finer than the sampling window. Pitches, balls, strikes, and the run itself arrive on a scale of seconds; our sampling collapses everything inside a thirty-one-second window into a single observation. A book that re-priced two seconds after the run and one that re-priced twenty-five seconds after it are recorded identically. This bounds the paper's ambition: no claim below the window is supportable, and we make none. The figure describes the apparatus, not the market.

This limit is worth stating plainly because it bounds the paper's ambition. Financial microstructure research often concerns latencies measured in microseconds. Nothing in this design speaks to that regime. What it can speak to is the slower, human-and-model-mediated process by which a sportsbook absorbs a publicly visible event, and whether two such processes can be distinguished from outside.

4. Data and institutional setting

4.1 The instrument

The data are generated by a continuously running collector that polls two sportsbooks and a game-state provider on a fixed schedule and appends every observation to an append-only panel. The properties below are verifiable facts about the apparatus rather than findings, and we state them as such.

Property	Value
Sampling cadence (live)	approximately 31 seconds, median, both books
Books observed	two, both retail-facing

Property	Value
Games with live quotes	in excess of one hundred, accumulating
Quote fields	posted line, both sides' prices, live flag, market status
Market status coverage	one book only
Game state	inning, half, outs, base occupancy, score, pitch count, times through order
Line semantics	full-game total, verified against pregame distribution

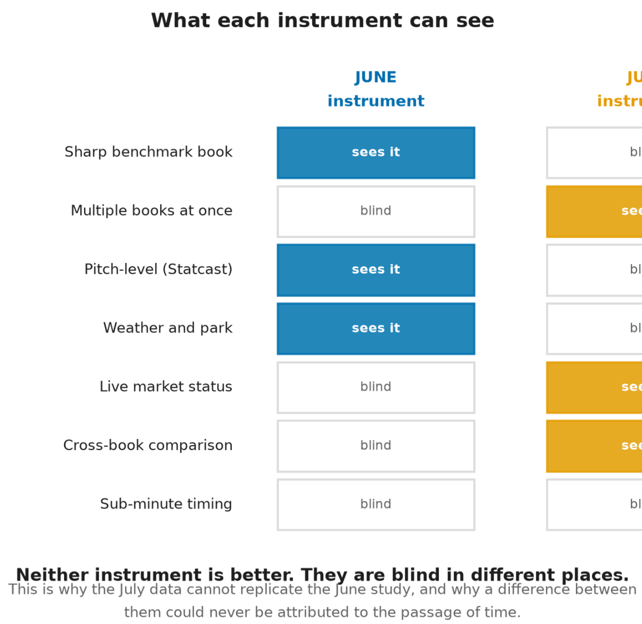

Figure 9. *Different instruments reveal different phenomena.* The June apparatus saw a sharp benchmark book, pitch-level measurement, and weather; it could not see two books at once or a market's live status. The July apparatus sees the second book and the status stream but is blind to everything the first could measure. Neither is better. They are blind in different places, which is why a result from one cannot confirm or contradict a result from the other.

4.2 What this instrument does not contain

Three absences are load-bearing for interpretation and are properties of the design rather than accidents of a particular sample.

There is **no sharp benchmark book**. Both observed books are retail-facing. The companion study used a sharp book as its incumbent forecast; nothing in this panel plays that role, and the leadership question here is therefore about two retail books rather than about a sharp-to-retail information cascade.

There are **no pitch-level or meteorological covariates**. The companion study's feature set is unavailable here.

Market status is asymmetric. One book publishes an explicit open, suspended, or removed state; the other publishes nothing. Any analysis that filters on tradeability can therefore only be applied to one book, and applying it to one and not the other introduces selection that favors the filtered book.

4.3 Why no comparison with the companion study is possible

It is tempting to treat this dataset as a second sample of the companion study's population. It is not. The two differ in the benchmark book, in the available covariates, and in the sampling cadence. Any difference in results between them would be jointly attributable to the passage of time and to the change of instrument, with no way to apportion between the two. We therefore make no claim in either direction about the companion study's conclusions. A temporal replication of that study requires re-running its own instrument, and is a separate exercise reported elsewhere.

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5. Methods

5.1 Constructing the event series

Events are extracted from the game-state panel as changes in the combined score between consecutive observations. The event time is the timestamp of the first observation exhibiting the new score, which is itself subject to the provider's own publication delay; A2 concerns exactly this, and the audit described in Section 5.6 quantifies it. Events are typed by magnitude, since a solo home run and a bases-clearing double are different information arrivals.

5.2 Constructing the price series

For each book, the main line is extracted per observation timestamp under a single fixed rule, and the series is reduced to the sequence of distinct quote states. Repeated identical quotes are not revisions and are collapsed. The extraction rule is fixed in advance, and the sensitivity of every reported quantity to that choice is reported alongside it, per A3.

5.3 Attribution

A revision is attributed to an event if it is the first distinct main-line change occurring within a fixed post-event window. The window is stated in the pre-registered plan below. Events whose windows overlap a subsequent event are flagged and analyzed separately, since attribution is ambiguous when two information arrivals are closer together than the response time being measured.

5.4 Estimation and inference

The unit of replication is the game, not the observation. Quotes within a game are strongly dependent, and treating them as independent produces intervals that are far too narrow. All estimation therefore proceeds by computing a per-game statistic and treating the game as the sampling unit, with intervals constructed accordingly.

5.5 Pre-registered analysis plan

The following is fixed before any estimate is computed. Departures from it will be reported as departures.

1. **Primary estimand.** The cross-book contrast in event-anchored response latency, `Δλ`.

2. **Event definition.** A change in combined score, typed by magnitude.
3. **Attribution window.** Three hundred seconds after the event. Events with a subsequent event inside the window are analyzed as a separate, flagged stratum.
4. **Main-line extraction.** A single odds-anchored rule, fixed and documented, with a mandatory sensitivity analysis across at least two alternative rules.
5. **Unit of replication.** The game.
6. **Primary comparison.** Direction and magnitude of `Δλ`, reported with a game-clustered interval.
7. **Mandatory falsification tests.** Before any leadership interpretation is offered: a symmetry test comparing the frequency of pre-event and post-event revisions, which separates a genuine response from a book's baseline revision rate; a shuffled-event placebo; and the extraction sensitivity analysis of item 4.
8. **Reporting rule.** A null result is reported with the same prominence as a positive one. No estimand, window, or extraction rule is added or altered after inspecting the results.

5.6 Conditions required before results may be reported

The Results section of this paper remains unwritten until all four of the following hold, each verified and recorded:

1. **A well-defined main line.** An extraction rule fixed in code and tested, with a demonstration that the primary statistic is materially invariant to reasonable alternatives.
2. **Clock comparability.** An audit establishing that event and quote timestamps are comparable, with residual skew quantified.
3. **Transport separability resolved in one direction.** Either an independent measurement of book-specific feed latency, or a defended argument that it is common-mode, or a documented demonstration that neither is achievable with this class of instrument. The third outcome satisfies this condition and converts the paper into a pure identification result.
4. **Robustness support.** A third live source, or an explicit outlier-detection procedure that does not require one.

Stating these in advance is the point. A paper that fixes its design before seeing its evidence cannot be reshaped by the evidence, and a null that arrives under those conditions carries the same weight as a finding.

5.7 Where this sits in the program

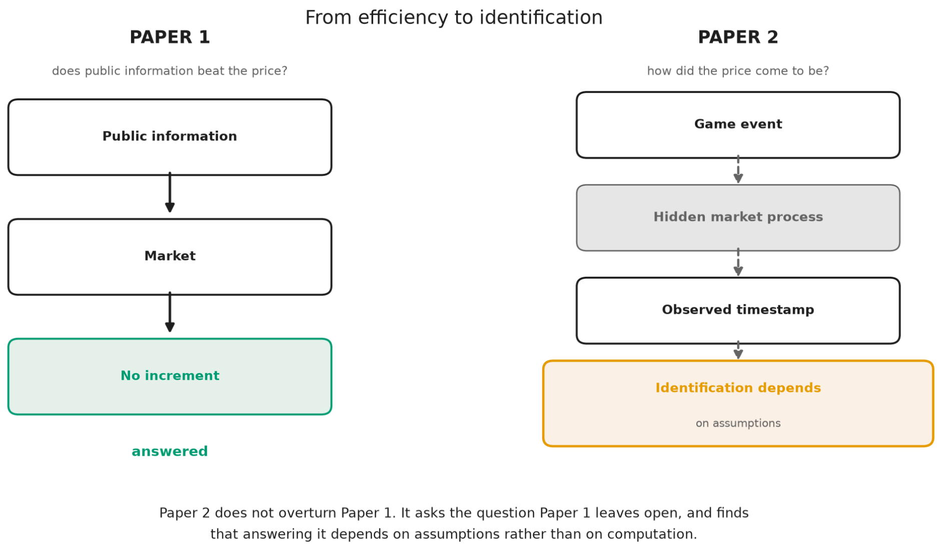

Figure 10. *Paper 2 does not overturn Paper 1; it asks what Paper 1 leaves open.* The companion study traced public information through the market to an answer, and every node it needed was observable. This study starts from an event, passes through a process it cannot see, and arrives at a timestamp whose interpretation depends on assumptions rather than on computation. The first paper ends in a finding. The second ends, at best, in a set of conditions.

That is not a weaker ambition. A literature that reports transmission estimates without establishing that transmission is identified accumulates numbers whose meaning depends on unexamined assumptions about publishing infrastructure. Stating the conditions is what makes the numbers interpretable when they eventually arrive.

Sections 6 (Results) and 7 (Discussion) are intentionally not drafted. See Section 5.6.